

Advanced (Carolina Reaper)

Q1. Factor the trinomial $x^2 + 5x + 6$

- (A) $(x + 2)(x + 3)$
- (B) $(x + 1)(x + 4)$
- (C) $(x - 2)(x - 3)$
- (D) $(x - 1)(x - 4)$
- (E) $(x + 2)(x - 3)$

Q2. Factor the trinomial $x^2 - 7x + 12$

- (A) $(x - 3)(x - 4)$
- (B) $(x + 3)(x + 4)$
- (C) $(x - 2)(x - 6)$
- (D) $(x + 2)(x + 6)$
- (E) $(x + 1)(x + 12)$

Q3. Factor the trinomial $x^2 + 2x - 15$

- (A) $(x + 5)(x - 3)$
- (B) $(x - 5)(x + 3)$
- (C) $(x + 2)(x - 7)$
- (D) $(x - 2)(x + 7)$
- (E) $(x + 1)(x - 15)$

Q4. Factor the trinomial $x^2 - 4x - 21$

- (A) $(x - 7)(x + 3)$
- (B) $(x + 7)(x - 3)$
- (C) $(x - 2)(x - 10)$
- (D) $(x + 2)(x + 10)$
- (E) $(x + 1)(x - 21)$

Q5. Factor the trinomial $x^2 + 10x + 24$

- (A) $(x + 4)(x + 6)$
- (B) $(x - 4)(x - 6)$
- (C) $(x + 2)(x + 12)$
- (D) $(x - 2)(x - 12)$
- (E) $(x + 1)(x + 24)$

Q6. Factor the trinomial $x^2 - 9x + 20$

- (A) $(x - 4)(x - 5)$
- (B) $(x + 4)(x + 5)$
- (C) $(x - 2)(x - 10)$
- (D) $(x + 2)(x + 10)$
- (E) $(x + 1)(x - 20)$

Q7. Factor the trinomial $x^2 + 6x + 8$

- (A) $(x + 2)(x + 4)$
- (B) $(x - 2)(x - 4)$
- (C) $(x + 1)(x + 8)$
- (D) $(x - 1)(x - 8)$
- (E) $(x + 3)(x + 3)$

Q8. Factor the trinomial $x^2 - 2x - 15$

- (A) $(x - 5)(x + 3)$
- (B) $(x + 5)(x - 3)$
- (C) $(x - 2)(x - 7)$
- (D) $(x + 2)(x + 7)$
- (E) $(x + 1)(x - 15)$

Open Ended Questions (FRQ)

Q9. Factor the trinomial $2x^2 + 5x - 3$ and explain your steps

Q10. Factor the trinomial $3x^2 - 2x - 5$ using two different methods and compare your results

Chef's Tips

- Tip 1: Encourage students to use different factoring methods
- Tip 2: Emphasize the importance of checking the results by multiplying the factors
- Tip 3: Make sure students provide clear explanations and steps for FRQs